There are two different types of upright drilling machines according to the cross-section of the column and they are

- Round column section upright drilling machine.
- Box column section upright drilling machine.

Base

Base is made of cast iron as it can withstand vibrations set by the cutting actions. It is mounted on the floor of the shop by means of bolts and nuts. It is supporting member as it supports column and other parts on it. The top of the base is accurately machined and has T-slots. When large work pieces are to be held, they are directly mounted on the base.

Column

Column stands vertically on the base and supports the work table and all driving mechanisms. It is designed to withstand the vibrations set up due to the cutting action at high speeds.

Work Table

Table is mounted on the column and can be adjusted up and down on it. It is provided with T-slots for work pieces to be mounted directly on it. Table may have the following adjustments.

Vertical adjustment obtained by the rack on the column and a pinion in the table.

Circular adjustment about its own axis.

After the required adjustments are made, the table is clamped in position.

Drill head

The drill head is mounted on the top of the column. It houses the driving and feeding mechanism of the spindle. The spindle can be provided with hand (or) power feed. There are separate hand wheels for quick hand feed and sensitive hand feed. The handle is spring loaded so that the drill spindle is released from the work when the drilling operation is over.

2.8 Radial Drilling machine

The radial drilling machine is intended for drilling and tapping on medium to large and heavy work pieces. It is used in mass production factories and to drill the holes at a diameter of 50mm and above.

A radial drilling machines is illustrated the given figure.

It consists of parts like base, vertical column, radial arm, Drill head. Elevating screw and driving mechanism of the spindle.

Base

The base is a large rectangular casting and mounted on the floor on the shop. Its top surface is accurately finished to support a column at one end and the table at the other end. T-slots are provided on it for clamping the work pieces.

Column

The column is a cylindrical casting which is mounted vertically at one end of the base. It supports the radial arm and allows it to slide up and down on its face. The vertical adjustment of the radial arm is effected by rotating a screw passing through a nut attached to the arm. An

XII BME_English version CHAPTER 02.indd 36 1/8/2020 1:44:51 PM